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Do Firms Proactively Build Internal Markets for Redeployment? Evidence from a Natural Experiment

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Abstract. The topic of related diversification is of fundamental interest to corporate strategy research. However, the understanding of how the potential for resource redeployment might encourage related diversification is limited. This paper explores the impact of the 2006 German real estate transfer tax (RETT) reform, which led to higher RETT rates but retained the exemption from RETT for corporate restructurings. We propose that firms might have reacted strategically to the threat of higher transfer taxes by combining related businesses into a portfolio, providing an efficient internal market for reallocating property assets in the event of business-specific shocks. Consistent with a dynamic perspective on relatedness, difference-in-differences analyses of the European retail market show that the probability of related diversification increased significantly for firms that were affected by the reform, relative to control firms. This finding contributes to the literature that links firm boundary choices to intertemporal economies of scope and offers new insights into the real effects of transfer tax policies.

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Keywords: corporate strategy • related diversification • resource redeployability • German real estate transfer tax reform

1. Introduction

Why do multibusiness firms exist given the advantages to focus and the costs associated with corporate headquarters? Rooted in Ansoff (1957), Penrose (1959), and Chandler (1962) and reinvigorated by Helfat and Eisenhardt (2004), one explanation gaining credence is that business portfolios enable firms to more flexibly pursue growth opportunities if their internal markets for nonfinancial resources are more efficient than external markets. Lieberman et al. (2017) predict that firms might proactively expand their scope to benefit from greater flexibility. Unfortunately, this prediction remains uninvestigated, largely because it is challenging to ascertain why firms diversify. In this paper, this prediction is tested through a natural experiment where firms change scope in response to expected increases in market frictions.

Our inquiry into whether firms intentionally expand horizontal scope to benefit from resource redeployability (i.e., the option to withdraw resources from one business and transfer them to another one) is most closely

related to two papers—one theoretical and one empirical. Lieberman et al. (2017) build on Dixit (1989) and Helfat and Eisenhardt (2004) to explain that market entry should be more likely for firms having resources that might be switched between current and new markets with low redeployment costs because redeployment potential reduces investment irreversibility and thereby, lowers entry thresholds (i.e., the expected performance required to induce entry). This prediction for scope expansion is unique from prior work emphasizing redeployment potential across an existing portfolio. Their emphasis on diversification into related markets derives from the belief that this is where redeployment costs are generally low because of the limited adjustments required to ready nonfinancial resources for new uses. This explanation for related diversification contrasts with explanations centered on the pursuit of synergy (i.e., intratemporal economies of scope). The natural experimental setting used in our study enables a focus on the redeployability explanation for related diversification. The second paper on which we build is

by Belenzon and Tsoimon (2016), who show that businesses in geographies with greater labor market frictions are more likely to be affiliated with a business group to benefit from internal labor markets. Rather than examining how broadly diversified business groups surmount labor market frictions, we consider how related diversified firms overcome real estate frictions burdening market exchange of retail stores.

Many types of frictions may inhibit market efficiency. In our empirical context of European retail chains, property transfer taxes, assessed when property changes ownership, may constitute up to 12.5% of property value. So, it should be no surprise that the prospect for higher property transfer taxes reduces market entry because they significantly increase the irreversibility of investment in retail stores (i.e., firms cannot costlessly adjust their stock of property assets in response to either positive or negative shocks to expansion opportunities at one business).¹ We consider that a different result might be obtained if market entry helps firms develop internal markets for property, and firms can transfer property assets between portfolio businesses with little adjustment (i.e., at low cost). Although several factors may drive internal redeployment costs, consistent with prior work, it is assumed that portfolio relatedness lowers redeployment costs (e.g., Helfat and Eisenhardt 2004, Sakhartov and Folta 2014, Lieberman et al. 2017, Sohl and Folta 2021). A key prediction is that firms become relatedly diversified to create an internal market for property, enabling them greater flexibility to take better advantage of growth opportunities or limit exposure to a business's poor prospects. Greater flexibility is obtained because the use of internal markets avoids property transfer tax payments, reducing investment irreversibility.

A natural experiment is used to test these ideas. In particular, the 2006 German real estate transfer tax (RETT) reform introduces a source of exogenous variation in the threat of higher transfer tax exposure, enabling a causal interpretation of how horizontal boundary choices are affected by expectations about increased market frictions. The reform empowered the 16 German federal states to set the RETT rate autonomously from 2007 onward while retaining the exemption from RETT for intragroup transactions in the process of corporate restructuring. The reform led to a substantial increase in the transfer tax rate across most German federal states around four years after the passage of the reform (see Table A1 in the Online Appendix). The focus of this paper is on firm responses to the *threat* of higher transfer taxes; it suggests that the outcome variable of interest—related diversification—enables an option to redeploy property assets at relatively low cost if transfer taxes increase in the future. The paper also clarifies that this benefit should accrue most robustly when company real estate portfolios contain larger proportions of owned property.

The retail market in the European Union (EU) was selected as the empirical context because real estate (i.e., retail stores) is a key asset in retailing and flexibility in adjusting the stock of real estate is of strategic importance for chain organizations. Difference-in-differences (DD) analyses of the 2006 German RETT reform provide causal evidence supportive of our prediction. The probability of related diversification increased by nine percentage points for treated firms in Germany in the years after the 2006 German RETT reform, relative to control firms in other EU countries. Several pieces of evidence give credence to the idea that related diversification is driven by a pursuit to obtain a potential for resource redeployment. For example, the treatment effect on related diversification is attenuated if firms operate in lease-intensive industry segments, supportive of the view that compared with owning, leasing might not provide the same level of discretion to redeploy and impetus to diversify before proposed transfer tax increases are implemented. The treatment has the opposite effect on unrelated diversification. Its effect on related diversification is accentuated when firms have larger real estate holdings, presumably because they might benefit from larger internal markets. The treatment also decreases market values but less so for related diversified firms. Although our natural experiment is specific to Germany, analysis reveals that the results generalize to other countries; increases in property transfer taxes raise the probability of related diversification across Europe, a finding unlikely to be explained by traditional arguments around intratemporal economies or “synergy.” Finally, anecdotes clarify that store redeployment occurs within firms that became relatedly diversified in Germany after 2006.

This paper contributes to two streams of literature. First, it extends an emerging stream explaining how redeployment flexibility might create value in an existing portfolio. It does so by investigating if firms diversify to generate redeployment flexibility. To illuminate the dynamic benefits of relatedness, we examine how exogenous variation in expectations about property transfer tax rates affects firm scope decisions, providing new insights into the evolution of horizontal firm boundaries. We also provide the first large-sample evidence around recent theoretical developments emphasizing the influence of redeployment potential on the dynamics of corporate diversification (e.g., Helfat and Eisenhardt 2004, Lieberman et al. 2017, Sakhartov 2017). A second contribution is in challenging the traditional view that market frictions, such as in the form of property transfer taxes, necessarily have a deleterious effect on entry decisions (e.g., Dachis et al. 2012, Lieser and Groh 2014, Devaney et al. 2017, Best and Kleven 2018). Our study considers the paradox that an expected increase in property transfer tax rates might encourage entry by established firms into related businesses. By exploiting variation generated by a sharp change in

transfer tax policy as a natural experiment, results show that under certain conditions, transfer taxes might increase firm scope rather than decrease it.

2. Background and Related Literature

Several bodies of literature bear upon the question of whether diversification is influenced by property transfer taxes. These literatures speak to how property transfer taxes increase investment frictions and how a firm's internal resource markets might provide a substitute for external market transactions.

2.1. Property Transfer Tax and Corporate Reorganization

Property transfer tax is a tax on real estate transactions with the selling price of the property as taxable base. Such taxes are common and raise substantial amounts of revenue in many countries (Best and Kleven 2018). The tax is determined by multiplying the property transaction price by the tax rate. By driving a wedge between the seller and buyer price, property transfer tax reduces asset values because taxes are capitalized into property prices, increasing capital costs to investors (Matheson 2012). Because property transfer taxes are imposed on each transaction, the volume of transactions decreases, which increases the irreversibility of property ownership. Dixit (1989) shows that greater investment irreversibility should make firms slower to enter and slower to exit in general. The literature on financial economics clarifies that transfer taxes are a major barrier to entry and investment, resulting in fewer transactions in the commercial and residential real estate market (e.g., Dachis et al. 2012, Devaney et al. 2017, Best and Kleven 2018). Indeed, it has been shown that firms extend the holding period of real estate properties in higher transfer tax environments—also known as the lock-in effect (Frenzel Baudisch and Dresselhaus 2018).

In recognition of a “fiscal unity” between a parent company and its business units or subsidiaries, most countries have laws for tax-free reorganizations, including the exemption of intragroup transfers of property assets from the payment of transfer tax (e.g., Vanistendael 1998). The objective of granting corporate reorganizations a tax-neutral treatment is that governments want to promote the continuity of firms and shareholder interests in the country. That is, the continuity of control within national boundaries is critical for governments in providing exemptions from transfer tax payments in the process of corporate reorganization (e.g., Jiménez-Valladolid de L'Hotellerie-Fallois 2014, PwC 2018).

2.2. Potential for Resource Redeployment

A recent spate of corporate strategy research emphasizes how internal markets for nonfinancial resources might substantially affect a firm's scope decisions (e.g., Anand and Singh 1997, Helfat and Eisenhardt 2004,

Levinthal and Wu 2010, Wu 2013, Sakhartov and Folta 2014, Lieberman et al. 2017, Sakhartov 2017, Santamaria 2022). It describes how the connection between relatedness and intertemporal economies of scope is unique from existing theories of corporate advantage (Folta et al. 2016), such as what Stein (2003) labels more-money effects (Lewellen 1971) and smarter-money effects (Stein 1997) in conglomerates. Levinthal and Wu (2010) emphasize that resource redeployment is pertinent for nonscale-free resources—such as physical resources like real estate—because they have capacity constraints, creating opportunity costs that require managers to choose how they should be allocated.²

By focusing on the comparative efficiency of alternative governance structures, such as markets and organizational hierarchy (Williamson 1987), resource redeployment decisions are linked to transaction costs. Several empirical studies show that redeployment benefits are fundamentally related to the presence of market frictions (e.g., Belenzon and Tzolmon 2016, Dickler and Folta 2020, Giarratana and Santaló 2020, Sohl and Folta 2021). In the absence of market frictions, when the price system reflects the full value of resources correctly accounting for opportunity costs, there would be no advantage of having the potential to internally redeploy resources. Even if market exchange of nonfinancial resources is costly, Helfat and Eisenhardt (2004) clarify that a firm can only profit from intertemporal economies of scope if internal redeployment costs do not offset the benefits of intertemporal resource transfer between businesses.

The most closely related research is that of Lieberman et al. (2017), who predict that the potential for resource redeployment increases the probability that firms will become related diversifiers. In their model, firms target related markets for entry because of lower sunk costs associated with such entries because resources can be redeployed relatively inexpensively to cope with changes in opportunities across businesses. Building on Dixit (1989), they reason that firms are encouraged to enter related opportunities at lower expected profits (than unrelated opportunities) because of greater reversibility of investment. In this sense, having the potential for resource redeployment is equivalent to a switching real option that lowers the expected performance required to induce entry.

2.3. Diversifying to Benefit from Internal Resource Markets

One expectation emanating from the literature is that holding all else equal, increases in external market frictions should make the prospect for internal markets more attractive and inspire related diversification to benefit from such markets. However, to our knowledge, no prior empirical work has examined the prediction that firms proactively pursue a related diversification

strategy to establish internal markets where nonfinancial resources can be redeployed. Indeed, this is the primary focus of our empirical effort, which is aided by a regulatory change in German real estate transfer tax affecting the market exchange costs for real estate—a firm-flexible (tradable) and nonscale-free resource. We believe that this shock, described in the next section, will help illuminate whether redeployment potential might influence related diversification.

A secondary focus is on clarifying that the resources at the heart of the redeployment decision need not be owned. Although ownership provides full discretion to redeploy, leasing might also facilitate redeployment, even if it does not offer the same level of discretion. Although leasing is a contractual governance structure, it may provide many of the same flexibility benefits for redeployment, provided that the lessee obtains contractual stipulations enabling discretion to redeploy.

3. The 2006 German Real Estate Transfer Tax Reform

Real estate transfer taxes are a common and important source of government revenues in many Organization for Economic Co-operation and Development (OECD) countries (Andrews et al. 2011). There is significant variation in tax rates across Europe, with Belgium imposing one of the highest tax rates on real estate transactions, whereas almost half of European member states have transfer tax rates below 5%. The German real estate transfer tax rate was set nationwide at 3.5% (on the sales price) between 1997 and 2006. In 2006, the RETT reform was passed by the German Parliament and dictated that each of the 16 German federal states could autonomously set the RETT rate after 2006.³ The reform ultimately led to a substantial increase in the RETT rate for 14 of 16 federal states in Germany by 2017 (see Table A1 in the Online Appendix). The average rate by 2017 was 5.4%. By 2011, the real estate transfer tax amounted to nearly 52% of the average property transaction costs (Fritzsche and Vandreï 2019). Additional details of this legislation and the corresponding increases are worth elaborating.

First, tax increases were not immediate. One state (i.e., Berlin) raised RETT from its base rate of 3.5% as early as 2007, one (i.e., Hamburg) raised RETT in 2009, one raised RETT in 2010, seven raised RETT in 2011, and six raised RETT in 2012. Ten states raised the RETT rate multiple times by 2017, including Berlin, which raised a second time in 2012 and a third time in 2014. In total, across the 16 states, there were 27 rate increases and no decreases between 2007 and 2017. So, the prospect of increased taxes was a real threat among retail firms throughout that 11-year postreform period, allowing for an extended time window to act before (further) increases raised the sunk entry cost.

Second, the RETT reform created incentives for German states to increase tax rates to better cope with the German federation system of fiscal equalization, where financial resources are redistributed across states to mitigate differences in fiscal capacity and expenditures (Buettner and Krause 2021). Because of the way fiscal capacity is defined by the German federation, higher tax rates have an adverse effect on fiscal capacity, which reflects the deadweight loss from taxation. Hence, a state that raises its tax rate receives more rather than fewer equalization transfers.⁴ The incentives to increase property tax rates were boosted further with a 2009 reform establishing state borrowing limits and imposing balanced budgets, effective 2020 (Fritzsche and Vandreï 2019). The RETT offered states an important means of achieving some revenue autonomy, with Christofzik et al. (2020) reporting that in 2019, RETT taxes constituted 61.1% of tax revenues captured at the state level, and it is the only tax where states can set their own tax rate.

Third, the anticipation of tax increases and the preannouncement of such increases led to a surge of property purchases just prior to implementation of tax increases. Within each state, changes to Germany's RETT rates are typically announced several months in advance because they must be approved by state parliaments. Christofzik et al. (2020) show that most tax increases were foreseeable (in press announcements) up to three quarters prior to their implementation. Fritzsche and Vandreï (2019) find that media citations of the tax reform are particularly high during the time that tax changes are announced or discussed by state parliaments and much less active around the actual date of tax increase. Accordingly, the timing of the tax changes is in most cases largely anticipated, with 46% more residential housing transactions in the months prior to a tax increase. Similar anticipatory effects were found in housing markets with changes to RETT in France (Poulhès 2020), the United Kingdom (Best and Kleven 2018), and Washington, DC (Slemrod et al. 2017). Only one study examines the anticipatory effects in commercial real estate markets, which are the focus of this paper, and finds some evidence that they exist one year prior to changes in German RETT rates (Frenzel Baudisch and Dresselhaus 2018). Overall, the evidence suggests that many market participants accelerate planned property purchases to take advantage of the lower tax rate in the period between announced and implemented transfer tax increases.

Fourth, subsequent to implementation of tax increases, property transaction volume dropped precipitously. Indeed, it is the lower volume of transactions after implementation of property taxes that describes the "lock-in" effect of transfer taxes because of how reduced liquidity increases sunk costs of property investment. Frenzel Baudisch and Dresselhaus (2018)

find that commercial property transactions in Germany drop with increases in property transfer taxes, whereas Lieser and Groh (2014) find the same effect in commercial real estate investments in 47 countries.⁵ An increase in the German RETT by 1% reduced single-family home transactions by 6% (Fritzsche and Vandrei 2019), whereas Frenzel Baudisch and Dresselhaus (2018) find that commercial property transaction volume declines by almost 12% with a 1.15% increase in taxes. The lock-in effect has also been observed with property tax increases in Toronto (Dachis et al. 2012); Washington, DC (Slemrod et al. 2017); Australia (Davidoff and Leigh 2013); and France (Poulhès 2020).

Fifth, subsequent to implementation of tax increases, it was not obvious whether leasing became relatively more attractive than purchasing. Prior research on the German RETT policy confirms that property values decline for commercial (Frenzel Baudisch and Dresselhaus 2018) and residential (e.g., Fritzsche and Vandrei 2019, Dolls et al. 2021, Koetter et al. 2022) properties when transfer taxes are implemented.⁶ The negative price reaction with property transfer tax increases is also robustly confirmed in locations outside of Germany (e.g., Dachis et al. 2012, Davidoff and Leigh 2013, Besley et al. 2014, Best and Kleven 2018). The impact on the average property price reflects the expectation that a given property will be subject to the tax each time it is sold, and thus, the expected incidence of the tax increase is capitalized into property prices. There is less evidence around the impact of RETT on lease rates. Although lessees are not subject to the property tax payment, lessors are. This suggests that landlords may compensate for their loss in investment (because of greater irreversibility) by increasing the burden on tenants (Frenzel Baudisch and Dresselhaus 2018, Lai and Milcheva 2023, Schneider and Wrede 2023).⁷ Higher rental rates should not be surprising because rent needs to cover all costs and yield a positive return for real estate lessors to be competitive among capital investments (Poterba 1984). Schneider and Wrede (2023) report that higher German real estate transfer taxes do not affect the probability of residential moves from renting to owning. Koetter et al. (2022) find that higher RETT rates lead to higher residential rental prices in urban areas of Germany but lower rent prices in rural areas. Studies of RETT changes in the United Kingdom confirm higher rental prices after transaction tax increases (Lai and Milcheva 2023). Given that property values and liquidity tend to decline and that lease rates tend to increase after transfer tax increases, it is ambiguous, *ex ante*, how the relative attractiveness of owning versus leasing is altered with transfer tax increases.⁸

Sixth, the reform did not affect the tax neutrality of real estate transactions within firms in the process of corporate restructuring. In particular, Section 6a of the German RETT Act provides for intragroup restructuring

transactions, an exemption from RETT payments. An important condition for the exemption to apply is that the restructuring transaction must involve one “controlling entity” (corporate parent) and one or more “controlled entities” (business units or subsidiaries). It also must involve a direct or indirect shareholding of at least 95% between the RETT group members for the five years immediately before and after the transaction (“prior holding” and “subsequent holding” periods). This five-year stipulation is consequential in that if firms competing in Germany expected that RETT would increase, they would benefit more by diversifying quickly to gain flexibility through an internal market for property.

This study examines whether firms exposed to the German RETT reform and therefore, the *threat* of higher transfer taxes were encouraged to become related diversifiers because of the higher relative benefits of internal resource markets. Having related businesses enables multibusiness firms to shift resources to businesses undergoing positive shocks and/or away from businesses undergoing negative shocks. Potential shocks to investment opportunities at one business can be either endogenous (caused by the firm) or exogenous (caused by factors outside of the control of the firm). It may be that one of a firm’s businesses will need capacity in property assets to fuel its growth, and internal redeployment is the more efficient alternative if internal redeployment costs are less than the transaction costs associated with the market. Presumably, even if a business is not performing poorly, there may be advantages of redeploying (part of) its property assets that have opportunity costs inside the firm to better-performing portfolio businesses. So, by combining related businesses in a portfolio, firms have the right, but not the obligation, to reallocate property assets at relatively low cost if there are inducements (i.e., performance differences between portfolio businesses) to do so. Firms can exercise this option in the future if business-specific shocks create positive inducements net of internal redeployment costs, which exceed the value that can be obtained using external market transactions.

The theory described suggests that the threat of higher transfer taxes may inspire related diversification to enable property redeployment and avoid sunk costs tied to higher taxes on future external transactions. Some elaboration is necessary given that, in retail, leasing is a common alternative to owning, and company real estate portfolios almost always contain some mixture of owned and leased real estate. These governance modes may differ with regard to (a) the discretion they allow to redeploy and (b) the impetus for diversifying in anticipation of transfer tax increases.

Complete discretion to redeploy should be available through property ownership (as described in Section 2.2), so the theory should apply with strongest force when ownership dominates a firm’s real estate

portfolio. For several reasons, leasing should also enable discretion to redeploy between leased and owned buildings or among leased buildings, even if it is at some reduced level relative to ownership. Redeployment discretion might be constrained or devalued if contracts allow for early termination by the landlord, allow for rent revisions, disallow rights of renewal, or disallow rights to sublease internally (i.e., between portfolio businesses). The German Civil Code disallows early lease termination by the landlord (other than a tenant breach), and the tenant may sublease with landlord consent (and can only be refused for good cause), although leases to the corporate parent (rather than one of its businesses) do not require subleasing to allow redeployment. Other leasing stipulations that may influence redeployment discretion might be determined by firm-specific negotiating power (Smith and Wakeman 1985), such as rights of renewal and rent revision.⁹ Greater discretion to redeploy via ownership should translate to greater flexibility to dynamically adapt property to portfolio businesses with better opportunities, and the value of such flexibility may offset any static efficiency advantage of leasing at a single point in time. Given that ownership may provide more discretion than leasing to redeploy and that transfer taxes have a direct bearing on the irreversibility of investment in owned property, we expect firms having property portfolios with larger proportions of owned property to be more likely to undertake related diversification prior to property transfer tax increases.

The impetus to diversify through property ownership should be especially pronounced if transfer taxes are expected to increase, so the diversifier can enable an internal property market without incurring the additional tax upon property acquisition. The impetus to diversify through leasing may be unaffected by expectations around transfer tax increases because it is unclear whether and when lease rates increase to capitalize future taxes on property transactions. However, because nearly all retail diversification contains some combination of owned and leased property, there should be some impetus for diversification prior to tax increases, with stronger impetus when the diversification involves a higher proportion of owned buildings. Although the focus of this paper is on related diversification, additional analysis reported in Section 6.4 also explores whether and how the German RETT reform affected the propensity for German retail firms to lease property.

Although firms may proactively diversify to create internal markets for property, it is worth elaborating available alternatives to related diversification beyond leasing in response to such a shock. The prospect of a transfer tax increase may inspire firms (a) seeking synergy to diversify sooner or (b) to diversify outside Germany. We also compare whether proposed transfer tax increases asymmetrically affect related versus unrelated

diversification. It is elaborated in Section 4 how unrelated diversifiers in our context lack potential to redeploy because it is extraordinarily costly to transform retail stores if portfolio chains use different store formats. The empirical analysis considers each alternative.

4. Empirical Design

4.1. Empirical Context

The EU retail market is used to test whether retail firms might have reacted strategically to the threat of higher transfer taxes by combining related businesses into a portfolio. The potential for redeployment of property assets is pertinent because many retail firms own multiple businesses or chains, whereas others are single-chain firms. The context is also attractive for testing the theory because it allows for a focus on the potential for redeployment of retail stores—a real asset of strategic and financial importance to multiunit organizations in many service industries, such as retail, hotel, restaurant, retail banking, and car rental (Garvin and Levesque 2008). Because redeployment of real estate is limited to portfolio businesses within a country, a focus on redeployment costs provides an important boundary condition on the potential for resource redeployment.¹⁰ Provided that redeployment can be made to a sibling chain within a country, firms still face redeployment costs (e.g., costs of changing a store's banner, interior design, and product assortment) to ready the real estate for use in an alternative chain. Although several factors may drive redeployment costs, lower redeployment costs should be obtained for retail stores in a country when there are related sibling chains in the country.

Business relatedness can be captured at the detailed submarket level within the retail industry by whether portfolio chains use the same store format. Store formats are commonly used in retailing to group similar chains based on physical properties, such as store size, location, and function (e.g., Gonzales-Benito et al. 2005, Cleeren et al. 2010, Corstjens and Lal 2012); they differentially characterize many different types of retail outlets, including but not limited to convenience stores, supermarkets, hypermarkets, department stores, and warehouse clubs. Consider, for example, the two Spain-based multichain clothing retailers “El Corte Inglés” and “Inditex.” El Corte Inglés operates multiple retail chains across store formats, such as the chains El Corte Inglés (department store format), Sfera (clothing store format), Opencor (convenience store format), Supercor (supermarket format), and Hipercor (hypermarket format). In contrast, Inditex operates eight chains (e.g., Zara, Berksha, Ohsho, and Pull & Bear) in the same clothing store format. Although both parent retailers likely benefit from the potential for resource sharing across retail chains (i.e., *intra-temporal* economies of scope), Inditex may have an additional advantage: the

option to switch retail stores at low cost between portfolio chains (i.e., *intertemporal* economies of scope).

The refined measurement of relatedness at the format level facilitates the study of redeployment costs of retail stores, which likely increase rapidly if businesses are less related, suggesting that redeployment potential should be influenced by whether portfolio chains adopt the same store format. For example, it would be economically unreasonable to transform a large hypermarket store into a small convenience store. In contrast, converting a department store from one department store chain to another will be economically feasible and more achievable because it does not require changes to a completely different store format, even if it might require adjustments in the form of signage and promotion. This is what Federated Department Stores did in 2001 when it closed its struggling Stern's department store chain in the United States and converted nearly 80% of the stores to its faster-growing Macy's and Bloomingdale's department store chains (Kaufman 2001). It follows that redeployment costs are relatively low if stores are converted for use by another chain in the same store format and that they should be excessively high if the alternative chain operates in a different store format. To capture these differences in redeployment potential, "same-format diversification" is defined as related diversification, and "different-format diversification" is defined as unrelated diversification. So, if El Corte Inglés was to move from one to two chains in a country, it would represent unrelated diversification, whereas a similar move by Inditex would represent related diversification.

Anecdotal evidence suggests that corporate parents recognize the potential for value creation by redeploying retail stores, both purchased and leased units, to their more productive chains if portfolio chains are related. During the coronavirus disease 2019 pandemic, for example, parent retailer Wesfarmers redeployed in Australia 92 stores from its struggling chain Target to its same-format sister chain Kmart while closing about 75 stores.¹¹ As a business reporter explained, Target has been "struggling for some time, really muscled out by its own stronger sister company, Kmart" (Wood 2020). Similarly, Foot Locker redeployed during the pandemic 18 stores of its Footaction chain to its better-performing same-format chains, such as Foot Locker and Champs Sports, while closing 32 Footaction stores (Foot Locker 2021). Chairman and chief executive officer (CEO) Johnson said on a call with analysts that "the company tested changing some Footaction stores into Foot Lockers and those locations became more productive as a result of the shift, which helped inform the decision to convert many of those stores" (Salpini 2021). In 2020, Foot Locker also decided to shut down its struggling Runners Point chain by redeploying several stores into other portfolio chains. For example, 72 Runners Point stores were closed in Germany, many of which were converted into Foot Locker

stores (HahnAG 2021). Another example is Delhaize Group, which redeployed in the United States 42 stores from its poorly performing Blooms chain to its better-performing Food Lion chain—both operating in the supermarket format while shutting down 7 stores.¹² In Australia, the United States, and Germany, corporate reorganizations are exempt from property transfer taxes if the ultimate ownership of the property remains the same (e.g., Gruidl and Kong 2012, Parkyn and Courtney 2016). Although anecdotal evidence suggests that parent retailers exploit internal markets for real property (stores), no systematic research examines if the creation of such internal markets might be inspired by the dynamic benefits of relatedness arising from redeployability.

4.2. Data

Unbalanced panel data of 955 publicly and privately held physical store-based retail firms in 27 EU countries during the eight-year window from 2003 to 2010 (three years before and four years after the 2006 German RETT reform) were obtained from Edge Retail Insight (formerly Planet Retail), totaling 6,083 firm-country-year observations.¹³ The data include retailers active in different industry segments, such as grocery, clothing and footwear, health and beauty, consumer electronics, leisure and entertainment, home and garden, and food service (i.e., restaurants and bars).

An advantage of these data relative to Compustat Segment data, which have been criticized because a firm's segments may not adequately reflect its actual business lines (e.g., Villalonga 2004), is that it provides a setting that allows for a refined measurement of business lines at the chain level. Besides providing information on retail firms and their business lines (chains) in a given country and year, the database also provides information on the store format and the number of stores of each chain. In particular, each chain operates multiple stores under the same banner (i.e., brand name) in a given store format, and none of the sampled chains changed their store format during the period studied.

Analysis is at the firm-country-year level. In the case of the France-based Carrefour Group, for example, the firm-country level corresponds to country divisions of Carrefour, such as Carrefour France and Carrefour Spain. Across these firm-country observations, the data illuminate whether Carrefour operates multiple chains (e.g., Carrefour City, Carrefour Express, Proxi, Carrefour Market, and Carrefour Hypermarket) and the store formats specific to the chains (e.g., convenience store, supermarket, and hypermarket).¹⁴

Table 1 provides summary statistics for the 6,083 firm-country-year observations. As shown in Table 1, 17.1% of observations are firms that are related diversified (i.e., operating at least two chains with the same format in a given country and year).¹⁵ Additionally, 6.5%

Table 1. Summary Statistics

Variable	Mean	Standard deviation	Min	Max
<i>Related diversified firm</i>	0.171	0.377	0.000	1.000
<i>Treated (firms in Germany)</i>	0.065	0.246	0.000	1.000
<i>Post (years after 2006)</i>	0.549	0.498	0.000	1.000
<i>Size (lnSales)</i>	18.713	2.489	9.762	24.696
<i>Years since country entry^a</i>	6.347	3.480	1.000	14.000
<i>Firm-country sales/firm sales</i>	0.148	0.298	0.000	1.000
<i>Parent public firm</i>	0.436	0.496	0.000	1.000
<i>Parent product diversification</i>	0.280	0.374	0.000	1.484
<i>Parent international diversification</i>	1.377	0.806	0.000	3.093
<i>GDP per capita (ln)</i>	10.190	0.646	8.416	11.630
<i>GDP per capita growth</i>	0.015	0.039	−0.145	0.130
<i>Inflation rate</i>	2.680	2.906	−9.654	23.153
<i>Internet users (per 100 people)</i>	58.847	18.356	8.900	91.000
<i>Market concentration</i>	0.071	0.048	0.023	0.667
<i>Tax burden</i>	0.605	0.135	0.298	0.895
<i>Investment freedom</i>	0.739	0.137	0.300	0.950
<i>Financial freedom</i>	0.715	0.144	0.400	0.900
<i>Monetary freedom</i>	0.813	0.048	0.508	0.908

Notes. The sample includes 6,083 firm-country-year observations from 955 retail firms in 27 EU countries for the period from 2003 to 2010. Table A3 in the Online Appendix presents correlations between these variables. Control variables are defined in Table A2 in the Online Appendix.

^aBecause Edge Retail Insight provides comprehensive information on country entry from 1997 onward, 1997 is used as the starting year for the *Years since country entry* variable if a firm entered before 1997.

of observations are 59 firms active in Germany (i.e., the treated group), meaning that 93.5% are firms active in the other 26 countries that were part of the EU during the sample period (i.e., the control group). Also, 54.9% of the observations occur after the 2006 reform.

4.3. Measurement Framework

Difference-in-differences analyses exploit the 2006 German real estate transfer tax reform to identify how expected changes in market frictions affect the probability of a firm undertaking related diversification in a country. A linear probability regression model is used, where the dependent variable *Related diversified firm* equals one if retail firm i operates (at least) two same-format chains in country c at year t and zero otherwise.¹⁶ A panel fixed effects (FEs) model is used with the following regression specification:

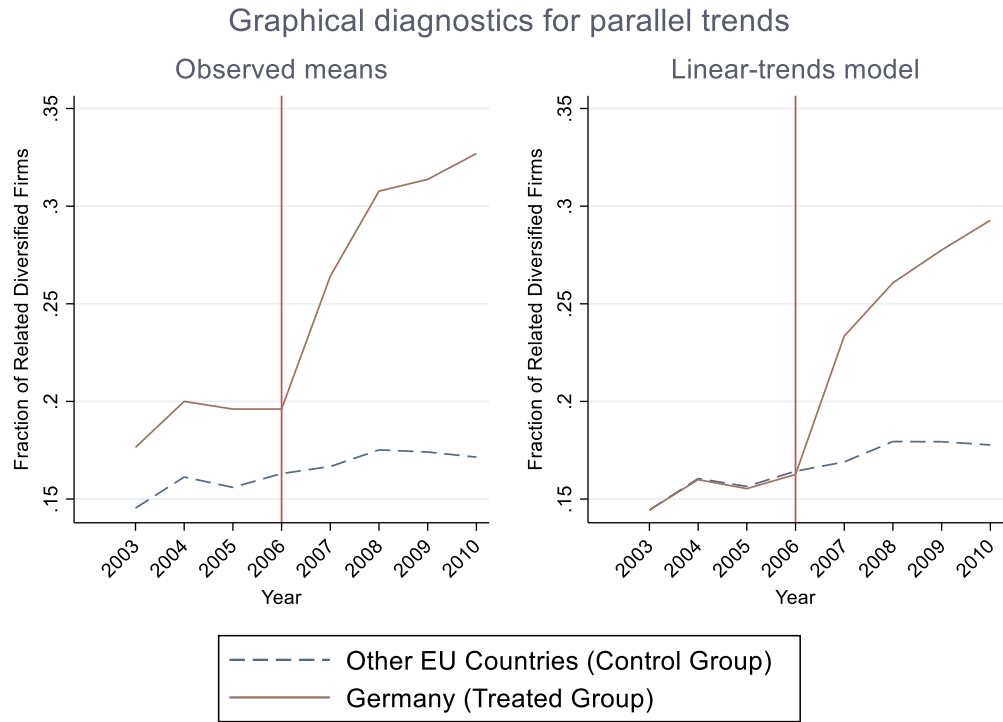
$$\begin{aligned} \Pr(\text{Related diversified firm}_{ict} = 1) \\ = \alpha_{ic} + \alpha_t + \delta \text{German firms postreform}_{ct} + \text{Controls}_{ict} \\ + \varepsilon_{ict}. \end{aligned} \quad (1)$$

Observations are treated if *German firms postreform* equals one, meaning that a firm is active in Germany after the reform between 2007 and 2010; *German firms postreform* equals zero if a firm is active outside of Germany or in Germany prior to 2007.¹⁷ So, δ is the coefficient of interest, capturing how the RETT reform influences the dependent variable, which should be a proxy for whether a firm proactively builds an internal market to enable the potential for real estate

redeployment. The specification also includes sets of firm-country fixed effects α_{ic} and year fixed effects α_t . Controls_{ict} denotes a vector of observed time-varying control variables defined in Table A2 in the Online Appendix. It aims to control for variation at the level of the firm (i.e., parent public firm, parent product diversification, and parent international diversification), the country (i.e., Gross Domestic Product (GDP) per capita, GDP per capita growth, inflation rate, internet users, market concentration, tax burden, investment freedom, financial freedom, and monetary freedom), and the firm-country (i.e., size, years since country entry, and firm-country sales/firm sales). Finally, ε_{ict} is an independent and identically distributed error term. Because the treatment effect is at the country level, standard errors are adjusted for clustering at the country level. Section 5.2 considers alternative ways of computing standard errors.

The DD estimator identifies a causal effect of the treatment by netting out trends from the control group. It does so by assuming that trends of related diversification in the control group represent trends in what would have happened in the treatment group in the absence of treatment. Analysis provides strong support for the parallel-trends assumption and a causal interpretation of the treatment effect. Figure 1 plots the means of the dependent variable over time for the treatment and control groups (the left panel of Figure 1). It also visualizes the results of the linear-trends model (the right panel of Figure 1), conditional on country and year effects. The trajectories of related diversification appear parallel for both groups prior to the 2006 German RETT reform but after the reform the lines diverge. The fraction of related diversified firms increased significantly in the treated

Figure 1. (Color online) The Fraction of Related Diversified Firms over Time



group after 2006, and this trajectory does not show any tendency toward returning to the levels prior to the 2006 RETT reform. A formal test for parallel trends in the pretreatment period, conditional on country and year effects, does not reject the null hypothesis of parallel trends ($F = 0.02$; $p = 0.89$).

Prior to running the model specified in Equation (1), lead and lag estimations were generated by interacting a German firm indicator variable (i.e., if the firm was active in Germany = 1) with year indicator variables in relation to the 2006 RETT reform. The dependent variable is *Related diversified firm*. Table 2 reports the results. The prereform coefficients are insignificant, suggesting that related diversification for German firms is similar to firms in other EU countries. However, the postreform coefficients are significant, suggesting that related diversification is significantly more likely for German firms than for their EU counterparts (Figure A1 in the Online Appendix illustrates the estimated coefficients and their 95% confidence intervals).

5. Results

5.1. Difference-in-Differences Estimates

Table 3 reports the main results, diagnosing the effect of the 2006 German RETT reform on related diversification. Columns (1) and (2) in Table 3 report cross-sectional regressions, and column (3) in Table 3 reports the panel FE regression specified in Equation (1). Different sets of fixed effects are added progressively to assess the

sensitivity of the reported results and mitigate concerns of omitted variable bias. Because Angrist and Pischke (2008) caution that the estimate of the treatment effect may be biased if poorly measured control variables are included, the specification reported in column (1) in Table 3 only includes country and year effects without controls. Results indicate that the estimated treatment effects are not driven by the choice of control variables.

Results across columns (1)–(3) in Table 3 show that the probability of related diversification increased significantly for firms in Germany after the 2006 German RETT reform relative to firms in other EU countries. All coefficient estimates are statistically significant at the 1% level. The estimated treatment effect is sizable; it lies between 9.0 and 9.2 percentage points. Conditional on country and year effects, results in column (1) in Table 3 indicate that in the years after 2006, the probability of related diversification increased in the treated group by 9.1 percentage points per year, on average, relative to the control group. Conditional on firm-country effects, year effects, and control variables, column (3) in Table 3 indicates that a firm in Germany had an 9.0-percentage-point increase in its probability of related diversification following the RETT reform relative to if the firm would not have been exposed to the reform. Overall, the results suggest a causal effect of the 2006 German RETT reform on decisions to pursue a related diversification strategy, consistent with a story around the dynamic benefits of relatedness arising from redeployability.

Table 2. Years Relative to the 2006 German Real Estate Transfer Tax Reform

Dependent variable: <i>Related diversified firm</i>	(1)	(2)	(3)
<i>RETT reform</i> (−2)	0.008 (0.010)	0.008 (0.011)	−0.003 (0.009)
<i>RETT reform</i> (−1)	0.008 (0.012)	0.004 (0.013)	−0.005 (0.010)
<i>RETT reform</i> (0)	−0.000 (0.013)	−0.002 (0.015)	−0.019 (0.012)
<i>RETT reform</i> (1)	0.063*** (0.014)	0.066*** (0.016)	0.057*** (0.011)
<i>RETT reform</i> (2)	0.097*** (0.012)	0.098*** (0.017)	0.086*** (0.015)
<i>RETT reform</i> (3)	0.103*** (0.014)	0.110*** (0.019)	0.090*** (0.016)
<i>RETT reform</i> (4)	0.117*** (0.016)	0.118*** (0.020)	0.111*** (0.018)
Control variables	No	Yes	Yes
Year effects	Yes	Yes	Yes
Country effects	Yes	Yes	No
Firm-country effects	No	No	Yes
Observations	6,083	6,083	6,083

Notes. Analysis is at the firm-country-year level. The data include 955 retail firms in 27 EU countries for the period from 2003 to 2010. This table reports the results of linear probability model regressions that examine leads and lags of the 2006 German real estate transfer tax reform. It shows the coefficients of the interaction terms between the *Treated* variable (i.e., firms in Germany) and time dummies with two leads (anticipatory effects) and four lags (posttreatment effects). The baseline year is 2003. Robust standard errors clustered at the country level appear in parentheses. The dependent variable *Related diversified firm* equals one if a focal retail firm operates (at least) two same-format chains in a given country and year and equals zero otherwise. Control variables are defined in Table A2 in the Online Appendix.

*** $p < 1\%$.

5.2. Robustness Checks

A number of robustness checks assess how confident we should be in results reported in Table 3. All are available upon request. A potential concern is that relatively few German firms moved from not relatedly diversified to relatedly diversified in the postreform period. To assess whether the small number of entry events by German firms after 2006 biases our results, several approaches were used. An initial approach examines whether the results are driven by any single firm by iteratively excluding each of the eight treated firms that became relatedly diversified after 2006 but were not relatedly diversified before the tax reform. This unreported analysis reveals that results are not driven by any single firm. To increase the number of portfolio reconfiguration events, a second approach alters the dependent variable to a firm’s number of related businesses in the given country and year. This raises the number of treated observations to 14 changes in the number of related businesses by firms in Germany after 2006. This change in the dependent variable yields positive and significant ($p < 1\%$) effects: 4.2- to 5.5-percentage-point increases in the probability that a German firm increases its number of same-format chains in any given year following the tax reform relative to control firms in other EU countries. The lower overall treatment effect using this more inclusive dependent

variable may also suggest only a marginal benefit to redeployability when having more than two related businesses in a portfolio. A final robustness check, given

Table 3. Results for the Probability of Related Diversification

Dependent variable: <i>Related diversified firm</i>	(1)	(2)	(3)
<i>German firms postreform</i>	0.091*** (0.008)	0.092*** (0.011)	0.090*** (0.011)
Control variables	No	Yes	Yes
Year effects	Yes	Yes	Yes
Country effects	Yes	Yes	No
Firm-country effects	No	No	Yes
Observations	6,083	6,083	6,083

Notes. Analysis is at the firm-country-year level. The data include 955 retail firms in 27 EU countries for the period from 2003 to 2010. This table reports the results of linear probability model regressions that examine the effect of the 2006 German RETT reform on the probability of related diversification using a difference-in-differences estimation strategy. Robust standard errors clustered at the country level appear in parentheses. The dependent variable *Related diversified firm* equals one if a focal retail firm operates (at least) two same-format chains in a given country and year and equals zero otherwise. *German firms postreform* is an indicator variable for firms in Germany after 2006. In total, 17.1% of the observations are related diversified firms. Control variables are defined in Table A2 in the Online Appendix. Table A4 in the Online Appendix reports results of control variables.

*** $p < 1\%$.

the small number of German diversifiers, examines patterns of related diversification in response to large increases in transfer tax in other EU countries between 2003 and 2010. This expands the number of initial related entries (i.e., when a firm moves from not relatedly diversified to relatedly diversified) to 26 entries after hikes in transfer taxes. This analysis is presented in Section 6.2 and shows that property transfer tax increases raise the probability of related diversification across Europe. These three approaches help dismiss concerns about sample size driving our results.

Additional robustness tests were conducted. All these tests produced substantively similar results to those reported in Table 3. First, as an alternative to including country and year fixed effects, the *Treated* (i.e., firms in Germany) and *Post* (i.e., years after 2006) variables were included (see Table A5 in the Online Appendix). Second, logit regression models were used instead of linear probability models (see Table A6 in the Online Appendix).¹⁸ Third, a more rigid DD analysis reported in Table A7 in the Online Appendix compares only observations for treated and control groups from the same parent firm (i.e., all parent retailers having no operations in Germany were dropped from this analysis). Fourth, segment-year joint fixed effects were included in all specifications (columns (1)–(3) in Table 3) to account for shocks during the sample period that were specific to a firm's industry segment measured at the two-digit Standard Industrial Classification (SIC) code level (the core segment was used if a firm operates in multiple segments). Fifth, given the relatively small number of groups in our setting (i.e., 27 EU countries), alternatives for computing standard errors were implemented using (a) clustering at the firm-country level rather than the country level (see Table A8 in the Online Appendix), (b) bias-corrected standard errors with the Bell and McCaffrey (2002) degrees-of-freedom adjustment, and (c) the aggregation method proposed by Donald and Lang (2007). Finally, to account for alternative macroeconomic and political factors potentially influencing our results, controls were added using interactions between the treated variable (i.e., firms in Germany) and country-level variables (i.e., GDP per capita, GDP per capita growth, inflation rate, internet users, market concentration, economic freedom indicators, and an employment protection index).¹⁹

In addition to the tests reported, a Granger causality test evaluated whether conditional on country and year effects, the treated and control groups changed their strategy of related diversification in anticipation of the 2006 German RETT reform. The null hypothesis of no effect in anticipation of treatment is not rejected ($F = 0.96$; $p = 0.43$), suggesting that firms did not change their related diversification behavior in anticipation of the reform. To further mitigate potential concerns that firms with certain characteristics might select countries in

anticipation of the 2006 German RETT reform, firms that entered a given country after 2003 were excluded from the sample; results were robust.

6. Additional Analyses

The analysis reported gives credence to the causal interpretation that firms implement related diversification prior to likely property transfer tax increases. Although this evidence is consistent with the idea that firms generate an internal real estate market to benefit from redeployability, the next subsections seek to further establish this theoretical interpretation as likely. This is done partly by driving at underlying mechanisms and partly by comparing alternative explanations.

6.1. Related Diversification vs. Unrelated Diversification

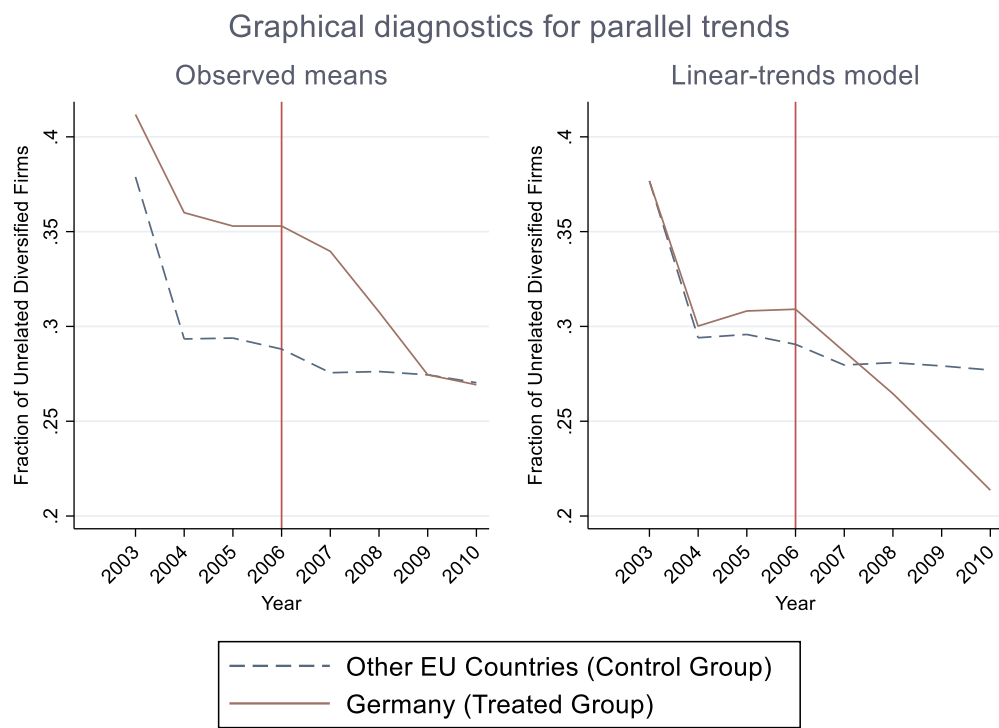
Analysis thus far shows how treatment affects related diversification. If the pursuit of internal real estate markets drives these effects, we would not expect analysis of unrelated diversification to produce the same effects. Unrelated diversification should be less sensitive to increases in real estate transfer taxes because store redeployment should be more costly across different store formats. The probability of diversification might be expected to decrease if firms do not plan to invoke internal markets because of higher sunk costs tied to real estate transactions after tax increases are implemented.

To test the prediction clarified, a dependent variable is created; *Unrelated diversified firm* equals one if a focal retail firm operates (at least) two different-format chains in a given country and year and equals zero otherwise. Figure 2 plots the means of the dependent variable over time and visualizes the results of the linear-trends model. Although the trajectories appear somewhat different between the treated and control groups in the pre-treatment period, results of the parallel-trends test do not reject the null hypothesis of parallel trends ($F = 2.79$; $p = 0.11$).²⁰ Columns (1)–(3) in Table 4 report a negative effect of the indicator variable for firms in Germany after 2006 on the probability of unrelated diversification. The negative relation between proposed transfer taxes and unrelated diversification reported contrasts sharply with our results showing increases in related diversification following the tax reform. This result is consistent with a redeployment story.²¹

6.2. Related Diversification Around the EU

The analysis was expanded to study whether property transfer tax changes influence related diversification across EU countries. This helps in several ways. First, the number of related entries (i.e., firms moving from not relatedly diversified to relatedly diversified) increases from 8 entries by firms in Germany after 2006 to 26 entries by firms in other EU countries after transfer

Figure 2. (Color online) The Fraction of Unrelated Diversified Firms over Time



tax hikes, reducing concerns about small numbers driving results. Second, it provides more generalizable evidence around our phenomenon of interest. Third, it helps dismiss concerns that other (i.e., nonreform) events in Germany around 2006 might explain related diversification.

Firms should be more likely to be related diversified in countries with higher property transfer taxes. A variable, *Property transfer tax*, is measured at the country-year level with the “tax rate on transfer of property” from the European Commission’s housing taxation database.²² Descriptive analyses provide some initial insight. Using a median split to distinguish between higher versus lower transfer tax environments, two-sample tests reveal that the proportion of related diversified firms is significantly different at the 1% level in high versus low transfer tax environments (20.1% versus 13.5%, respectively), whereas the proportion of unrelated diversified firms is similar in both tax environments (approximately 29%).

To examine whether firms are more likely to build an internal market for real estate in response to noticeable increases in property transfer tax in a given country, an indicator variable was created, where *Transfer tax hike* equals one if a tax increase from the prior to the current year falls in the top 5% of the changes in property transfer tax distribution. We also explore whether noticeable decreases in property transfer tax might influence the probability of related diversification, where *Transfer tax cut* equals one if a tax decrease falls in the bottom 5% of

the changes in property transfer tax distribution. It is noteworthy that German firms are “control” firms in this analysis. Multivariate regressions are presented in Table A9 in the Online Appendix. Columns (1)–(5) in Table A9 in the Online Appendix report effects on related diversification, showing that all coefficient

Table 4. Results for the Probability of Unrelated Diversification

Dependent variable:			
<i>Unrelated diversified firm</i>	(1)	(2)	(3)
<i>German firms postreform</i>	−0.038*** (0.006)	−0.020** (0.009)	−0.021*** (0.007)
Control variables	No	Yes	Yes
Year effects	Yes	Yes	Yes
Country effects	Yes	Yes	No
Firm-country effects	No	No	Yes
Observations	6,083	6,083	6,083

Notes. Analysis is at the firm-country-year level. The data include 955 retail firms in 27 EU countries for the period from 2003 to 2010. This table reports the results of linear probability model regressions that examine the effect of the 2006 German RETT reform on the probability of unrelated diversification using a difference-in-differences estimation strategy. Robust standard errors clustered at the country level appear in parentheses. The dependent variable *Unrelated diversified firm* equals one if a focal retail firm operates (at least) two different-format chains in a given country and year and equals zero otherwise. *German firms postreform* is an indicator variable for firms in Germany after 2006. In total, 29.2% of the observations are unrelated diversified firms. Control variables are defined in Table A2 in the Online Appendix.

p < 5%; *p < 1%.

estimates of the *Transfer tax hike* variable are positive and statistically significant at the 5% level. For example, column (5) in Table A9 in the Online Appendix indicates that a hike in a country's transfer tax rate (i.e., a tax increase of more than 3%) enhances the probability of related diversification by a firm in that country by 1.7 percentage points. Note that these effects are lower than the effects reported in Table 3, but this analysis studies actual tax increases, whereas earlier analysis studied proposed (but not yet enacted) tax increases, where the impetus to diversify should be higher. Results also reveal insignificant effects of the *Transfer tax cut* variable, perhaps because the potential for synergy is unaffected by the efficiency of real estate markets, and firms might remain relatedly diversified if transfer taxes decrease to avoid the loss of synergy. Columns (6)–(10) in Table A9 in the Online Appendix show that the probability of unrelated diversification is not significantly affected by changes in transfer tax, consistent with our explanations in Section 6.1. By expanding our analysis beyond Germany, these findings provide additional support of the view that higher market frictions caused by transfer tax enhance the relative advantage of related diversifiers.

6.3. Lease- vs. Ownership-Intense Industry Segments

The following analysis explores the possibility that property leasing versus owning may bear upon a firm's discretion to redeploy and its impetus for diversifying prior to a tax increase. This analysis exploits differences in lease ratios across 10 retail industry segments as illustrated in Table B1 in Online Appendix B, where lease ratio is the ratio of the capitalized operating lease expenditure to the sum of that amount and the amount of property, plant, and equipment (PP&E) (Beatty et al. 2010).²³ For example, segments consisting of food stores (SIC 54), eating and drinking establishments (SIC 58), general merchandise stores (SIC 53), and apparel and accessory stores (SIC 56) have median lease ratios of 42.8%, 53.6%, 62.9%, and 85.4%, respectively. To examine whether the response to the tax reform is weaker among retail firms in lease-intense industry segments, an indicator variable is created, where *Lease-intense segment* equals one if the median lease ratio of a firm's industry segment exceeds the retail median lease ratio and zero otherwise. Cross-sectional variation in the treatment effect is examined by including interactions between the treatment variable (i.e., *German firms postreform*) and *Lease-intense segment*. Table 5 reports the results. Columns (1) and (2) in Table 5 show that results from the main analysis remain when *Lease-intense segment* is included. Columns (3)–(5) in Table 5 show negative and significant coefficients for the interaction between the *German firms postreform* and *Lease-intense segment* variables, indicating that the effect of the RETT

reform on the probability of related diversification is attenuated for firms in lease-intense segments. Coefficient estimates (i.e., column (4) in Table 5) show that following the tax reform, the probability of related diversification increases by 13.0 percentage points for retailers in ownership-intense industry segments, whereas it increases by 3.1 percentage points for retailers in lease-intense industry segments relative to control firms. The difference is statistically significant at the 1% level.²⁴ It is noteworthy that the tax reform increased the probability of related diversification for firms in both own-intense and lease-intense segments.

Additional analysis also considered only the 191 public firms in the sample with necessary data on leasing to examine interactions with firm-level lease intensities (see Table A10 in the Online Appendix). Although the results of this analysis fall short of a causal interpretation, they are consistent with the idea that related diversification decisions by firms more oriented toward leasing are less sensitive to changes in transfer tax rates.

6.4. Lease Intensity Following the Tax Reform

One might argue, for several reasons, that the German RETT reform increases the relative attractiveness of leasing versus owning. First, higher property transfer taxes increase the sunkness of property ownership. Second, leasing may allow discretion to redeploy while avoiding sunk investment in property. Section 3 argued that, ex ante, it is ambiguous how property transfer taxes should affect the relative attractiveness of leasing versus owning, but here, we empirically test whether firm-level lease intensities increased subsequent to the tax reform.

Two additional data sets were obtained (see Online Appendix B for details). The first is a sample of all public retail firms in EU countries with lease ratio information included in the Compustat Global database. An advantage of the Compustat Global analysis is that it allows a comparison between treated firms (i.e., firms listed on public exchange in Germany) with control firms (i.e., firms listed on public exchanges in other EU countries). Table B2 in the Online Appendix reports results using this sample and *Lease ratio* as the dependent variable (i.e., the ratio of a firm's capitalized operating lease payment to the sum of that amount and the amount of PP&E). There is a negative association ($p < 1\%$) between the indicator variable for firms in Germany after 2006 and the lease ratio variable. Column (3) in Table B2 in the Online Appendix indicates that conditional on firm effects, year effects, and control variables, subsequent to the RETT reform a German firm's lease ratio decreased by 6.9 percentage points, on average, relative to control firms in other EU countries.

The second is a proprietary sample from the Corporate Balance Sheets (Ustan) 1987–2019 data set (Becker et al. 2020), which was provided by the Research Data and Service Centre of the Deutsche Bundesbank (for the

Table 5. Interactions with Segment-Level Lease Intensity

Dependent variable: <i>Related diversified firm</i>	(1)	(2)	(3)	(4)	(5)
<i>German firms postreform</i> (<i>Treated</i> × <i>Post 2006</i>)	0.092*** (0.008)	0.093*** (0.011)	0.129*** (0.012)	0.130*** (0.014)	
<i>Lease-intense segment</i>	−0.088*** (0.021)	−0.062*** (0.019)	−0.088*** (0.020)		
<i>Lease-intense segment</i> × <i>Treated</i>			−0.014 (0.020)		
<i>Lease-intense segment</i> × <i>Post 2006</i>			0.053*** (0.016)	0.056*** (0.017)	0.055*** (0.018)
<i>German firms postreform</i> × <i>Lease-intense segment</i>			−0.079*** (0.017)	−0.099*** (0.018)	−0.101*** (0.019)
Control variables	No	Yes	Yes	Yes	Yes
Year effects	Yes	Yes	Yes	Yes	Yes
Country effects	Yes	Yes	Yes	No	No
Firm-country effects	No	No	No	Yes	Yes
Country-year effects	No	No	No	No	Yes
Observations	6,083	6,083	6,083	6,083	6,083

Notes. Analysis is at the firm-country-year level. The data include 955 retail firms in 27 EU countries for the period from 2003 to 2010. This table reports the results of linear probability model regressions that examine the interaction effect between the 2006 German RETT reform and operating in a lease-intense industry segment on the probability of related diversification using a difference-in-differences estimation strategy. Robust standard errors clustered at the country level appear in parentheses. The dependent variable *Related diversified firm* equals one if a focal retail firm operates (at least) two same-format chains in a given country and year and equals zero otherwise. *German firms postreform* is an indicator variable for firms in Germany after 2006. *Treated* is an indicator variable for firms in Germany. *Post 2006* is an indicator variable for years after 2006. *Lease-intense segment* is an indicator variable that equals one if the median lease ratio of a firm's industry segment is above the retail sector's median lease ratio (i.e., 64.7%), where lease ratio is the ratio of the capitalized lease expenditure over the sum of PP&E and capitalized lease expenditure (Beatty et al. 2010). The initial (i.e., core) segment is used if a firm operates in multiple segments. Summary statistics of the lease ratio variable by industry segment are reported in Table B1 in Online Appendix B. The measure of *Lease-intense segment* is identical if only the prereform period is used to compute (segment and sector) median lease ratios. In total, 39.8% of the observations are firms in lease-intensive industry segments. Control variables are defined in Table A2 in the Online Appendix.

*** $p < 1\%$.

project number 2023\0026 with on-site data access: GaFo). It includes public and private retail firms in Germany with information on whether company buildings are predominantly leased (one if yes). An advantage of the Bundesbank data is the comprehensiveness of retail firms in Germany (i.e., 1,502 firms). Table B3 in the Online Appendix shows that firm-level lease intensities were not significantly different between the postreform period (2007–2010) and the prereform period (2003–2006). Table B4 in the Online Appendix examines how actual transfer tax changes affect lease intensities. Tax increases are negatively associated with the probability that German retailers predominantly lease buildings ($p < 5\%$). The analysis in this section does not lend support to the view that leasing becomes relatively more attractive with either proposed or implemented property transfer tax increases.

6.5. Synergy

Although results are consistent with a redeployment story, it is possible that they may also be explained by the fact that proposed but not yet enacted tax increases inspire earlier entry by firms that were otherwise planning to diversify to take advantage of synergy. Four issues lead us to believe that our related entrants are not

pursuing synergy. First, Levinthal and Wu (2010) emphasize that synergy is most practical for scale-free resources, and real estate is a nonscale-free resource. Second, anecdotal evidence reported in Section 6.10 illuminates that redeployment of real estate occurred subsequent to related entry by German firms in the postreform period. Third, analysis presented in Section 6.2 shows a significant relationship between noticeable property transfer tax increases and related diversification across the EU, irrespective of any proposed increases providing a window to expand before increases accrue. We cannot think of any reason why related diversification to pursue synergy would be positively related to property transfer tax increases. A fourth issue leading us to believe that synergy is not driving the results is elaborated in Section 6.6.

6.6. Company Managed vs. Franchised Stores

Assumptions about flexibility to redeploy may depend on whether chain stores are company managed or franchised; redeploying stores between chains might be more feasible and less costly when they are company managed rather than franchisee operated. Because Holmstrom and Roberts (1998) argue that restaurant chains often use business-concept franchising, whereas

other chain stores, such as supermarkets, clothing stores, and department stores, are typically company managed, an indicator variable was created for restaurant firms to capture a firm’s focus on franchising in the given country.²⁵ Table A11 in the Online Appendix shows a negative and significant interaction effect between the *German firms postreform* and *Restaurant firm* variables ($p < 1\%$), which would be expected if firms respond less to the threat of higher transfer taxes by becoming relatedly diversified when their business model relies more on franchising. To elaborate, the coefficient estimates in column (4) in Table A11 in the Online Appendix indicate that the probability of related diversification increased by only 0.7 percentage points for restaurant firms, whereas it increased by 10.3 percentage points for nonrestaurant firms. If restaurant firms have less potential for store redeployment but similar potential for synergy (e.g., via knowledge sharing across chains) as compared with other retail firms, this finding provides evidence supportive of the view that redeployability, rather than synergy, inspires related diversification in response to the RETT reform.

6.7. Real Estate Portfolio Size

If the proposed effects around redeployability are correct, then firms more likely to pursue related diversification should be those operating more real estate assets.

This is because firms should be more likely to use an internal market more extensively if a larger number of stores is subject to potential redeployment. Table 6 provides evidence supportive of this view, showing a positive and significant interaction ($p < 1\%$) between the *German firms postreform* variable and an indicator variable that equals one if a firm-country’s number of stores is above the sample median in the year before the tax reform on the probability of related diversification.²⁶ To elaborate, column (4) in Table 6 shows that following the tax reform, the probability of related diversification by a firm in Germany increases by 2.7 percentage points if the firm operates few stores, whereas it increases by 10.3 percentage points if the firm operates many stores in the year before the tax reform relative to firms in other EU countries.

6.8. Geographic Diversification

It is likely that the prospect for property transfer tax increases in Germany spurred firms to invest in other countries rather than Germany. This does not preclude our explanation that German tax increases will enhance the relative advantage of related diversifiers in Germany.²⁷ Indeed, unreported analysis at the firm-year level using *Sales percentage in Germany* as the dependent variable and including firm effects and parent firm controls (size, public firm, product diversification,

Table 6. Interactions with Real Estate Portfolio Size

Dependent variable: <i>Related diversified firm</i>	(1)	(2)	(3)	(4)	(5)
<i>German firms postreform</i> (<i>Treated</i> × <i>Post</i> 2006)	0.083*** (0.007)	0.090*** (0.011)	0.040*** (0.014)	0.027** (0.011)	
<i>Large real estate portfolio</i>	0.214*** (0.023)	0.139*** (0.025)	0.135*** (0.023)		
<i>Large real estate portfolio</i> × <i>Treated</i>			0.013 (0.032)		
<i>Large real estate portfolio</i> × <i>Post</i> 2006			0.002 (0.015)	0.004 (0.014)	−0.006 (0.015)
<i>German firms postreform</i> × <i>Large real estate portfolio</i>			0.065*** (0.015)	0.076*** (0.012)	0.084*** (0.013)
Control variables	No	Yes	Yes	Yes	Yes
Year effects	Yes	Yes	Yes	Yes	Yes
Country effects	Yes	Yes	Yes	No	No
Firm-country effects	No	No	No	Yes	Yes
Country-year effects	No	No	No	No	Yes
Observations	6,083	6,083	6,083	6,083	6,083

Notes. Analysis is at the firm-country-year level. The data include 955 retail firms in 27 EU countries for the period from 2003 to 2010. This table reports the results of linear probability model regressions that examine the interaction effect between the 2006 German RETT reform and real estate portfolio size on the probability of related diversification using a difference-in-differences estimation strategy. Robust standard errors clustered at the country level appear in parentheses. The dependent variable *Related diversified firm* equals one if a focal retail firm operates (at least) two same-format chains in a given country and year and equals zero otherwise. *German firms postreform* is an indicator variable for firms in Germany after 2006. *Treated* is an indicator variable for firms in Germany. *Post* 2006 is an indicator variable for years after 2006. *Large real estate portfolio* is an indicator variable that equals one (between 2003 and 2010) if a firm-country’s number of stores in 2005 is above the sample median number of stores in 2005 (i.e., 42 stores) and equals zero otherwise (because the tax reform might affect a treated firm’s real estate portfolio size, relative to control firms, the year 2005 is used to determine real estate portfolio size). In total, 47.3% of the observations are firms with large real estate portfolios in 2005. Control variables are described in Table A2 in the Online Appendix.

** $p < 5\%$; *** $p < 1\%$.

international diversification, GDP per capita, and GDP per capita growth) showed that the sales percentage that a firm generates in Germany decreased by 0.8 percentage points in the postreform period relative to the prereform period ($p < 1\%$).

6.9. Firm Value Tied to Related Diversification

It is possible the stock market recognizes the potential extra value creation opportunities through redeployment potential within related diversifiers. Two additional analyses are used to test this proposition. The first examination uses an event study methodology to exploit the stock market reaction to the approval of the reform by the German Bundesrat on July 7, 2006 and how shareholder reactions might be influenced by whether firms are relatedly diversified. The stock market reaction is captured by the cumulative abnormal return (CAR) during an “event window” computed by using the market model and portfolio approach (Brown and Warner 1985). Because the event occurs for all firms on the same day, we correct for cross-sectional correlation of abnormal returns. The sample, which consists of 17 public retail firms that are listed on the German stock exchange in 2006, and the calculation of CAR are described in detail in Online Appendix C. Table C1 in the Online Appendix reports the results. It shows that results for the portfolio of related diversified firms are significant on the event day (0) and over the two-day event windows including the event day ($-1, 0$) and ($0, 1$). For example, on the event day (i.e., July 7, 2006), the CAR is 4.4% for the portfolio of related diversifiers. The CARs of related diversified firms in the two-day intervals excluding the event day (i.e., $(-2, -1)$ and $(1, 2)$) are substantively smaller and insignificant. Results are also insignificant during all event windows for a portfolio consisting of unrelated diversified and focused firms. These findings suggest that the approval of the reform by the Bundesrat had a positive influence on the portfolio return of related diversified firms but had no influence on the portfolio return of unrelated diversified and focused firms.²⁸

A second examination considers a broader sample of both treated and control firms between 2003 and 2010 using panel data analysis. The sample is described in Online Appendix C. It includes 43 firms (17 treated firms that are listed on the German stock exchange and 26 control firms that are listed on stock exchanges in other EU countries) for the analysis of market-to-book values and 42 firms (17 treated firms and 25 control firms) for the analysis of excess returns. Table C2 in the Online Appendix reports the results using *Market-to-book value* and *Excess return* as the dependent variables.²⁹ Column (1) in Table C2 in the Online Appendix indicates that in the years following the 2006 German RETT reform, market-to-book values of retail firms listed on public exchange in Germany (these were all German

firms) declined by 1.5%, on average, relative to retail firms listed on public exchanges in other EU countries ($p < 1\%$). Results also reveal that the negative association between the RETT reform and the market-to-book values of German firms is attenuated for related diversified firms, as indicated by the significant and positive interaction effect between the *German firms postreform* and *Related diversified firm* variables ($p < 1\%$). To elaborate, column (4) in Table C2 in the Online Appendix shows that following the tax reform, the market-to-book value of a German firm decreases by 1.6% if the firm is unrelatedly diversified or focused on a single business, whereas it decreases by only 0.6% if the firm is relatedly diversified relative to control firms in other EU countries. Using *Excess return* as the dependent variable produces substantively similar results (see columns (5)–(8) in Table C2 in the Online Appendix). These findings suggest that the stock market anticipated that the relative advantage of related diversifiers should increase following the 2006 German RETT reform.

6.10. Did Firms Redeploy Stores After 2006?

The focus of this paper is on the probability of firms becoming related diversifiers to generate internal markets, where these firms have an option but not an obligation to redeploy in the future. Our analysis focused on the establishment of this option. Of course, having the option to redeploy does not imply that redeployment will occur. Neither does a lack of redeployment detract from the value of having the potential. Nevertheless, the proposed mechanism is further illuminated if postexpansion redeployment sometimes obtains within our sampled firms. Anecdotal evidence indicates that it does. For example, in 2007, German-based Do it yourself (DIY) retailer Globus acquired Hela in the same DIY format with 31 stores in Germany. During 2019 and 2020, Globus redeployed six Hela stores to Globus Baumarkt in a corporate restructuring process. Furthermore, in 2007, another German-based DIY retailer Praktiker acquired Max Bahr in the same DIY format with 77 stores in Germany. In 2012, Praktiker started a corporate restructuring program with the redeployment of 57 Praktiker stores to the more profitable sister chain Max Bahr until 2013. In 2013, the United States-based Foot Locker added in Germany the chain Runners Point. During 2020 and 2021, all 72 Runners Point stores were closed in Germany, and many of them were converted into Foot Locker stores (HahnAG 2021). Lastly, in 2020, the German-based shoe retailer Deichmann abandoned its midprice brand Roland in the same shoe store format including 45 stores after about 32 years of operation. By the end of 2019, 20 stores were transferred to the more profitable portfolio chain Görtz Schuhe, and they were quickly rebranded as Görtz (HahnAG 2020).

Interestingly, firms redeployed both purchased and leased stores, with some retailers, like Praktiker,

redeploying mainly leased units.³⁰ To complement these direct observations, we searched for large-sample evidence of store redeployment from exiting to surviving portfolio chains as found by Sohl and Folta (2021). If store redeployment occurs and our theory around redeployment costs is correct, upon a chain's country exit, we should observe more store growth in surviving related sibling chains than in surviving unrelated sibling chains. The 2008 global financial crisis (GFC) is used as an environmental shock—a period during which exit via internal redeployment likely became more pronounced because external markets were less efficient. Countries were hit by the GFC at different times, and the systemic banking crisis database (Laeven and Valencia 2013) was used to identify this timing.

Unreported analysis of 1,663 chains in all 27 EU countries showed that if a focal country was hit by the GFC, a surviving chain's store growth increased by 2.0 percentage points with each additional store of an exiting same-format sibling ($p < 1\%$), conditional on chain-country effects and control variables. In contrast, the increase in store growth was only 0.2 percentage points in noncrisis times ($p < 5\%$), suggesting that more firms exit via divestment (i.e., by selling stores in the external market) in more stable economic times. Moreover, results revealed that exits of different-format siblings were not significantly associated with a focal chain's store growth, whether the chain's country was hit by the GFC or not. These large-sample patterns solidify our view that, in our context, related diversification is a necessary condition for exercising the option to redeploy stores if conditions warrant.

7. Discussion

A spate of recent research clarifies how and when multi-business firms might benefit from internal (noncapital) resource markets, and this research is beginning to provide empirical evidence on these issues. Prior to this paper, however, it has been unknown whether firms proactively diversify to generate these internal markets. We take advantage of an anticipated German real estate transfer tax increase, which would eventually raise the irreversibility of real estate investment, to provide evidence that retail firms move quickly to benefit from the corporate flexibility inherent in more efficient internal markets for real estate. Several theoretical and empirical contributions are worth elaborating.

First, we speak to the dynamics of diversification. An evolving literature on resource redeployment has largely focused on benefits subsequent to diversification but has been relatively silent around whether these benefits inspire diversification or whether myopic managers merely redeploy within the portfolio they inherit. Like Lieberman et al. (2017), we draw attention to the fact that firms should be more inclined to pursue related

diversification because lower irreversibility of investment lowers the required expected profit to induce entry. We clarify that this inclination should escalate with greater external market inefficiency. In this sense, we explain how diversification decisions are linked to the relative efficiency in internal versus external markets for nonfinancial resources.

Second, we are able to focus on redeployability as an explanation for related diversification. This is surely because of our empirical context in the European retail market and choice of natural experiment. Because retail stores have the characteristics of a firm-flexible, nonscale-free resource (i.e., they are tradable in the open market and have opportunity cost in their usage), the threat of higher market exchange costs for retail stores and the exemption from such costs for internal reallocations should be largely tied to diversification decisions inspired by the potential for store redeployment. We find compelling evidence that rates of related diversification increase in Germany relative to other EU countries, and we interpret this as firms diversifying to create internal resource markets to substitute for anticipated decreases in external market efficiency. Adding to our confidence in this conclusion is that (a) several firms that became related diversifiers in response to the treatment exercised the option to redeploy stores, (b) the legislation does not inspire unrelated diversification, (c) firms focusing on franchising are less sensitive to the policy reform, (d) firms with larger real estate holdings are more likely to pursue related diversification, and (e) there is a strong relationship between actual property transfer tax increases and related diversification across the EU.

Third, we have argued that both owning and leasing real estate should enable the potential to redeploy. We know of no other studies that consider whether redeployment potential is available through leasing. This is important because retail firms nearly always use a combination of both. Evidence is consistent with the view that firms with either high or low lease-intensive real estate portfolios are more likely to pursue related diversification if transfer taxes are expected to increase. The fact that more lease-intensive firms are less likely to pursue related diversification lends credence to the idea that compared with owning, leasing does not provide the same level of discretion to redeploy and impetus to diversify quickly. Because we do not observe to what extent related diversification occurs through leasing or owning, it may be possible that diversification through leasing is preferred to owning, but our analysis of retail firms suggests that lease ratios do not increase subsequent to reform implementation.

Fourth, our work relates to prior research on internal capital markets. In particular, the theoretical analysis of Stein (1997) focuses on redeployment of financial resources, where the key advantage of the multibusiness

firm is informational. Managers have unique insight into opportunities, and this insight is accentuated for related diversified firms. Our internal market story contains those elements, but it is accentuated when a firm's nonfinancial resources lack market liquidity and can be internally redeployed at some cost. We believe that our natural experiment of the German RETT reform complements prior research on the allocation of financial resources because it has a direct bearing upon the liquidity of property, a key resource in retail.

Finally, our research reconciles with but also extends a large literature in financial economics examining the impact of transfer taxes on transactions in the real estate market (e.g., Dachis et al. 2012, Lieser and Groh 2014, Devaney et al. 2017, Best and Kleven 2018). This literature uniformly predicts that increases in transfer taxes will decrease investment in property. We show that this need not necessarily be obtained in the short term if the possibility of transfer taxes encourages the creation of an internal market for property as an alternative to the external market. By considering the paradox of higher transfer tax rates and more diversifying entry into business lines at the fine-grained level of submarket entry, our study provides novel insights into the real effects of transfer tax policies.

Our paper has several limitations, which provide directions for future research. Given our large-scale empirical effort to test how the potential for redeployment affects related diversification, it is important to note that we do not empirically distinguish between the potential for resource sharing and resource redeployment as determinants of related diversification decisions, and we search for evidence consistent with our theoretical prediction. In particular, we try to overcome this by testing theoretical conditions and choosing a context where the dynamic benefits of relatedness are the most plausible explanation of decisions to combine related businesses into a portfolio.

We do not directly observe redeployment beyond anecdotal evidence. Indeed, it is not our intent to observe redeployment as it exemplifies the exercise of the switching option and has been examined elsewhere (e.g., Belenzon and Tsoimon 2016, Sohl and Folta 2021). Rather, our focus is on the initiation of such an option.

A potential concern is generalizability. As with many interesting policy experiments, a question is whether the findings are informative beyond the 2006 German RETT reform as a source of market frictions and our focus on the potential for redeployment of real estate (stores) in the retail industry. Although Lieberman et al. (2017) expect that the potential for resource redeployment might inspire diversifying entry, their interviews suggest that managers do not consider flexibility when entering. We believe that our focus on the retail industry may be particularly appropriate for diagnosing farsightedness in anticipating redeployment because our

interviews and anecdotal evidence suggest store redeployment is common for related diversified firms. One executive at Inditex suggests that redeployment occurs regularly. Although our findings should certainly apply to other service industries in which multiunit chains are a prevalent organizational form, we believe that the framework described in this paper is more generally applicable across various sources of market frictions and industry contexts. As just one example, firms in the airline industry often operate multiple businesses (i.e., airlines) in parallel. If buying and selling or licensing airplanes become more costly because of market frictions, it is possible that switching airplanes between portfolio businesses creates value if there are inducements net of redeployment cost to do so. It is also worth emphasizing that our dichotomous distinction between “related” and “unrelated” diversification is specific to our retail context, where store format is a fundamental determinant of redeployment costs. Other contexts may usher in more subtle increments of redeployment costs, requiring a less discrete treatment of types of diversification.

Location and competition are certainly important issues in retail. We fully expect that competitive preemption plays an important role in decisions to create an internal market for retail stores to the extent that firms may prefer to acquire or hold onto valuable properties, both owned and leased, by having the option to internally redeploy them. Informal discussions with executives at Inditex suggest that this is a common occurrence. Future research might pursue this line of inquiry: for example, by examining how transfer taxes and considerations of competitive preemption might jointly affect firm scope decisions.

Overall, by emphasizing the dynamic benefits of relatedness arising from resource redeployability, this paper contributes to an improved understanding of the influence of relatedness on diversification decisions. It, therefore, sheds new light on a fundamental question in strategy research of what value the corporate parent adds and the consequent implications for firm boundary choices (Rumelt et al. 1994).

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Endnotes

¹ Both positive and negative shocks to expansion opportunities are relevant in this context because it is largely irrelevant whether the buyer or seller pays the transfer tax to the government.

² On the other hand, scale-free resources are used for contemporaneous resource sharing.

³ On July 7, 2006, the German Bundesrat approved the “federalism reform,” and the changes came into effect on September 1, 2006.

⁴ Fritzsche and Vandrei (2019) report that right-wing governments implemented fewer RETT increases than left-wing governments.

⁵ A rise in the mean tax rate in their sample (i.e., 4.14%) by the average tax rate increase (i.e., 1.15%) reduces office transaction volume by almost 12% and other commercial buildings by almost 5%.

⁶ Dolls et al. (2021) found that a one-percentage-point increase in the German tax rate reduces property prices by an average of 3%. Some work suggests small and insignificant effects on prices (e.g., Petkova and Weichenrieder 2017, Slemrod et al. 2017).

⁷ Several German reports on the 2006 RETT reform also indicate that landlords “passed on” transfer tax increases to tenants to compensate for their increased investment irreversibility (e.g., Rappen 2011, Scherf and Dresselhaus 2016).

⁸ The analysis reported in Section 6.4 explores this question.

⁹ The latter may influence the payoff to redeployment if rent is revised when the building is reallocated across a firm’s portfolio businesses (i.e., retail chains).

¹⁰ As such, our focus is different from studies such as Kogut and Kulatilaka (1994) and Berry (2010), which show that multinational firms have flexibility in switching production across countries depending on relative production cost across countries. We also emphasize business relatedness as a determinant of the potential for redeployment of real estate (i.e., retail stores).

¹¹ Note that despite the similar logo, name, and format, Target Australia has no relation to the American discount retailer Target Corporation.

¹² According to Pierre-Olivier Becker, former CEO of Delhaize Group, “This decision is in line with our ‘New Game Plan’ which is aimed at accelerating profitable growth ... We feel these decisions are in keeping with our responsibility to our shareholders to deploy resources where they will achieve the highest return” (Delhaize 2012).

¹³ The sample includes all physical store-based retailers that were part of the top 350 global firms and operated at least one chain in at least 1 of the sampled 27 EU countries between 2003 and 2010.

¹⁴ For example, “Carrefour City” and “Carrefour Express” are two different chains in the same convenience store format. According to its 2023 report, Carrefour’s retail chains operated in Europe a real estate portfolio consisting of 6,605 stores in the convenience store format, 2,869 stores in the supermarket format, and 703 stores in the hypermarket format.

¹⁵ In total, 29.2% are unrelated diversified firms (i.e., operating at least two chains, all having different formats in a given country and year), and 64.4% are firms with a single chain in a given country and year.

¹⁶ All entry events into new chains by treated firms after 2006 include open market transactions, where the acquired retail store portfolio contains both owned and leased properties.

¹⁷ To our knowledge, Germany’s 2006 RETT reform was unique during this period, with the following exceptions. Greece imposed value-added tax on new buildings in 2006, and France amended the tax treatment of the French source dividends paid to European companies in 2007 (PwC 2018). Analysis excluding observations in Greece and France (from the control group) produced consistent results. Similarly, results were consistent when observations from the EU countries that implemented large transfer tax increases between 2003 and 2010 (see Section 6.2) were excluded from the control group.

¹⁸ We used a linear probability model in the main analysis (and a logit model in robustness checks) because we were interested in whether a firm is more likely to pursue related diversification subsequent to the tax reform but not in the specific timing of a diversification event. In unreported analysis, we also estimated a discrete-time proportional hazards model. In this analysis, the dependent variable *Related diversified firm* equals one only in the year of related entry and zero otherwise. Once a firm has entered a second same-format chain in a given country, it was removed from further consideration. The hazard analysis produced consistent results (however, we could not address issues of left censoring because of insufficient observations of country entries during our sample period).

¹⁹ The employment protection index was obtained from the OECD employment database and is also used by Belenзон and Tsolmon (2016) in their study on labor market frictions and business group formation. The employment protection index was available for 21 EU countries, reducing our sample to 5,189 observations.

²⁰ Figure A2 in the Online Appendix plots the estimated coefficients from the leads and lags exercise.

²¹ Substantively similar results were produced with a multinomial logit model, indicating that German firms postreform were more likely to be relatedly diversified relative to either focused (i.e., single-chain firms) or unrelatedly diversified ($p < 1\%$).

²² In unreported analyses, we included the *Property transfer tax* variable in addition to the other variables of the specifications reported in Table 3. Results were robust, suggesting that our findings are not influenced by the tax rate on the transfer of property in other EU countries (i.e., the control group).

²³ See Online Appendix B1.2 for a detailed description of the lease ratio variable.

²⁴ The specification reported in column (5) in Table 5 adds country $\times$ year joint fixed effects. This allows treated and control groups to follow different trends. The coefficient estimate on *German firms postreform* $\times$ *Lease-intense segment* remains significant and comparable in magnitude with the estimate in column (4) in Table 5.

²⁵ Restaurants in our sample include firms such as Burger King, Costa Coffee, McDonalds, Starbucks, Subway, and Yum Brands, which all focus on business-format franchising. The decisions to use franchising and leasing are independent as indicated in Table B1 in the Online Appendix, where the median lease ratio for restaurants (53.6%) is below the median of the retail sector (64.7%).

²⁶ A potential concern is that the tax reform might affect a treated firm’s real estate portfolio size relative to control firms. To address this issue, the year 2005 is used to determine a firm’s real estate portfolio size. Using the year 2003 or 2004 (or the prereform average) to determine real estate portfolio size or total amount of square meters instead of the number of stores as a measure of real estate portfolio size produces substantively similar results.

²⁷ Presumably, even if a firm seeks to avoid any further expansion into new stores in Germany, it could still generate an internal market by entering a related business via redeployment of current stores.

²⁸ Interestingly, an unreported analysis showed insignificant results for both stock portfolios (i.e., related diversifiers versus unrelated

diversifiers and focused firms) and all event times if September 1, 2006—the day when the RETT reform changes came into effect—was used as the event day. This suggests that shareholders reacted positively when firms were relatedly diversified at the time that the RETT reform was approved rather than at the time when the reform came into effect, which would be expected if abnormal returns reflect new information that arrives in the market.

²⁹ *Market-to-book value* is the ratio between the market value of the firm and the book value of its assets (e.g., Berger and Ofek 1995, Morandi Stagni et al. 2020). A firm's market value equals the sum of its market capitalization and book value of debt. Using the market-to-sales value as the dependent variable produced qualitatively similar results. *Excess return* is the difference between a firm's stock return and a risk-free rate (i.e., the return rate from Germany's short-term Federal Treasury notes) at the end of the fiscal year.

³⁰ Internal transfer of leases might have been beneficial not only because it avoids costs of external subleasing to third parties (e.g., search and renegotiation costs) or early lease termination fees but also, because it avoids costs of new lease negotiations for the faster-growing sister chains.

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